

Lecture 9

Calculating with Probability

Bayesian Networks

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1

Today's lecture is about calculating with probabilities and how/when we use probability in AI.

Overview

- 1 Calculating with Probability
- 2 Applying Probability
- 3 Bayesian Networks (PGMs)

Part 0: Basics of Probability

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3

Let's review the basic notions of probability.

Sample Space

The **sample space** of an experiment or trial is a set of all possible outcomes.



$$\Omega = \{\text{heads, tails}\}$$



$$\Omega = \{1, 2, 3, 4, 5, 6\}$$



$$\Omega = \{(1;1), (1;2), (2;1), \dots, (6;6)\}$$

The elements in the set are called sample points, possible worlds, or **atomic events**.

The **sample space** is all the possible outcomes of an experiment, denoted by an omega symbol. For a coin toss, the sample space is only heads or tails because these are the only single outcomes of this experiment. Every single element in this set is called an **atomic event**. In an experiment involving the rolling of two dice, each atomic event is the combination of outcomes of these two dice, that is, every atomic event is a pair.

The Event and Probability Spaces



Event Space $E = \{\{\}, \{1\}, \{2\}, \dots, \{1,2\}, \dots, \{1,2,3,4,5,6\}\}$

Events are...

1. subsets of the sample space (and the things we assign probability to).
2. the atoms in our language.

Unless the sample space is infinite, its powerset $P(\Omega)$ is usually considered the event space.

A **probability space** is a sample space with an assignment $P(w)$ for every atomic event $w \in \Omega$, and event A

- $0 \leq P(w) \leq 1$
- $P(w_1) + P(w_2) + \dots + P(w_n) = 1$ for all $w_i \in \Omega$
- $P(A) = P(w_1) + P(w_2) + \dots + P(w_n)$ for all $w_i \in A$

Note: $\{1,2\}$ stands for either 1 or 2 happened. This is different from $(1;2)$ which we had before.

Events are the atoms or building blocks of our probability language. Events can be combinations of atomic events or any other subset of the sample space. For a sample space that is finite, meaning there is not an infinite number of possible atomic events, the powerset is considered the event space. Because a powerset always include the empty set, the empty set is also part of the event space.

We can transform a sample space into a probability space by assigning probabilities to each event. The probability value is always between 0 and 1, 0 meaning it never happens and 1 meaning it always happen. The probability for all atomic events sum up to 1, because it is always true that one of the atomic events happens. The probability of a complex event A is defined as the sum of all possible atomic events of the experiment.

Examples of Spaces

Possible outcomes of rolling a die can be represented by sample space
 $\Omega = \{1,2,3,4,5,6\}$

Examples of events

$A_1 = \{2,4,6\}$ “outcome is even”

$A_2 = \{1,2,3,4\}$ “outcome is smaller than 5”

$A_3 = \{6\}$ “outcome is 6”



Example of a probability function

$P(w) = 1/6$ (for each atomic event, i.e. 1, 2, 3, etc.)

Then $P(\{6\}) = 1/6$, $P(\{2,4,6\}) = 1/2$, and $P(\{1,2,3,4\}) = 2/3$

6

An event w can be an atomic event itself (e.g. outcome is 6) or a condition (e.g. outcome is even, outcome is smaller than 5). We can assign a probability value to these events $P(w)$ by dividing the number of atomic events in the event by the length of the sample space *if* each atomic event is equally likely. In a dice roll, where each number is equally likely, we can find the chance that the outcome is smaller than 5 by dividing $4/6 = 2/3$ because there are 4 atomic events in which the outcome is smaller than 5 out of 6 possible atomic events.

Random Variables

A **random variable** is a function from atomic events to some **range**, like T/F, real numbers, or discrete values{sunny,rainy}.

e.g. Odd(1) = true, Odd(2) = false, Weather = rainy

P induces a probability distribution $\mathbf{P}(X)$ for all variable values of X

$$P(X = x) = P(w_1) + P(w_2) + \dots \text{ for a random variable } X \text{ and atomic events } w_i \\ \text{where } X(w_i) = x$$

$P(X=x)$ is the probability of an event where the random variable X has value x , and is just the sum of all the atomic events with that property.

Example $P(\text{Odd}=\text{true}) = P(1) + P(3) + P(5) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$

$\mathbf{P}(\text{Odd}) = (\frac{1}{2}, \frac{1}{2})$ as probability distribution for values true and false.

7

A random variables is a function that gives an atomic event a particular range. This range can be binary (e.g. true or false), real numbers (e.g. 1, 1.2), or discrete values (e.g. cold, hot). Given the random variable “odd” with range {true, false}, we can say that Odd(1) is true because 1 is an odd number, and Odd(2) is false. Probability P induces a probability distribution for all the random variables. If we want to calculate the probability that the random variable Odd is true, we sum all the individual events to find that there is a $\frac{1}{2}$ probability that a given value is Odd.

$P(X=x)$ is the probability of an event where the random variable X has value x , and is just the sum of all the atomic events with that property.

Language Extension for Boolean Random Variables

A **Boolean proposition** is the event where the proposition is assumed to be true unless negated. If we have Boolean random variables A and B , we can abbreviate:

- Event a = set of atomic events w where
 - $A(w)=\text{True}$
- Event $\neg a$ = set of atomic events w where
 - $A(w)=\text{False}$
- Event $a \wedge b$ = set of atomic events e and f where
 - $A(e)=\text{True}$ and $B(f)=\text{True}$
- Event $a \vee b$ = set of atomic events e and f where
 - $A(e)=\text{True}$ or $B(f)=\text{True}$

8

For Boolean random variables denoted with capital letters (A), we assume that the proposition is true unless negated. This allows us to abbreviate the language such that a means that random variable A is true for w , $\neg a$ means A is false, $a \wedge b$ means A and B are true for e and f , and $a \vee b$ means either A or B (or both) are true.

Syntax for Propositions

	Boolean	Discrete	Continuous
Random Variable Range	Odd true/false	Card <diamond,heart,spade,club>	Value 10, 4
Proposition	Odd = true or <i>odd</i>	Card = diamond	Value = 10 Value < 4

For Boolean combination of basic propositions ($\wedge$, $\vee$, $-$, etc.) with related probabilities

$$P(a \vee b) = P(a) + P(b) - P(a \wedge b)$$

e.g. $P(\text{odd} \vee \text{Value} > 4) = P(\text{odd}) + P(\text{Value} > 4) = \frac{1}{2} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6}$

Otherwise written $P(\{1,3,5,6\}) = \frac{4}{6}$

For Boolean, discrete, and continuous variables we denote **propositions**, or true statements, with = to indicate the true value of the variable. For Boolean propositions as shown on the last slide, we can use symbols of propositional logic to combine these statements in formulas.

Part 1: Calculating with Probability

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10

Let's combine and use these basic notions.

Probability Distribution

A **probability distribution** (notation bold **P**) gives probability values for all possible assignments of a Random Variable.

e.g. probability distribution at the beginning of the game, with *heart* as Trump for the Random variable Card and range <diamond, heart, club, spade> is:

$$\mathbf{P}(\text{Card}) = \langle \frac{5}{19}, \frac{4}{19}, \frac{5}{19}, \frac{5}{19} \rangle$$

For a given Random Variable , the probability distribution gives the probability of all the values in the Range.

11

A probability distribution **P** has values for all possible assignment or outcomes of an event. In a game of Schnapsen where the Trump card has been removed from the deck we have $\mathbf{P}(\text{Card}) = \langle 5/19, 4/19, 5/19, 5/19 \rangle$. There is an equal probability for pulling each type of card except for hearts, because there are now less hearts in the deck. There are also only 19 possible cards to still pull because one heart has been removed.

Joint Probability Distribution

Joint probability distribution shows a probability distribution for two or more random variables co-occurring, *e.g.* for the random variables of card type and value>4:

$P(\text{Card}, \text{Value}>4)$ is a 4x2 matrix of values:

Value	Card	diamonds	hearts	spades	clubs
Value>4=True		1/9 $10\spadesuit$	2/9 $As\heartsuit$ $10\heartsuit$	0	2/9 $11\clubsuit$ $10\clubsuit$
Value>4=False		0	2/9 $Q\heartsuit$ $K\heartsuit$	0	2/9: $K\clubsuit$ $Q\clubsuit$

after only 9 cards $As\heartsuit$ $10\heartsuit$ $Q\heartsuit$ $K\clubsuit$ $K\heartsuit$ $11\clubsuit$ $10\clubsuit$ $Q\clubsuit$ $10\spadesuit$ are left unknown and with Trump card $J\heartsuit$ known.

$$P(\text{diamond}, \text{value}>4) = 1/9$$

$$P(\text{Card}, \text{value}>4=F) = \langle 0, 2/9, 0, 2/9 \rangle$$

$$P(\text{heart}) = 4/9$$

$$P(\text{heart}, \text{value}>4) = \langle 2/9, 2/9 \rangle$$

$$P(\text{value}>4=F) = 4/9$$

12

A **joint probability distribution** gives the probability for two or more random variables occurring together. Let's combine the probability distribution we just saw for Card type with the probability of pulling a card with a value greater than 4. Recall that only the 10 and the Ace have a value greater than 4. In a scenario with 9 cards left unknown and $J<3$ as the Trump card, we have a 1/9 probability of pulling a diamond valued greater than 4, and a 0 probability of pulling a diamond valued less than 4. From this table we can read off value for probabilities of specific events P , or all the probabilities that belong to a particular distribution P .

Probability Distribution *Example*

A probabilistic view on passing the IS course (frequentist/Bayesianist):

WorkGroup	Lecture	Pass	Frequency	Probability
F	F	F	200	0.2
F	F	T	50	0.05
F	T	F	50	0.05
F	T	T	50	0.05
T	F	F	50	0.05
T	F	T	150	0.15
T	T	F	50	0.05
T	T	T	400	0.4

$P(\text{passing the course}) = P(\text{Pass}=\text{T}) = 0.05 + 0.05 + 0.15 + 0.4 = 0.65$

$P(\text{attending lecture}) = P(\text{Lecture}=\text{T}) = 0.05 + 0.05 + 0.05 + 0.4 = 0.55$

These are called the Marginal Probabilities. Why?

13

The frequencies in the table come from observations made in previous years of teaching this course. The frequency is the number of students for whom the conditions on the left were true, for example, 200 students do not attend lectures or workgroups and fail the course. Because the total frequency of students adds up to 1000 students, the probability of a student belonging to each group is the frequency with the decimal point moved three times to the left.

To calculate the total probability of a random student passing the course, all the probabilities for which Pass=T need to be summed. We can also sum the probabilities in the rows where Lecture=T to find how many students attend the lectures.

Marginal Probabilities

The **marginal probability** is the probability of a single event occurring, independent of other events.

Probability Distribution *Example Revisited*

WorkGroup	Lecture	Pass	Frequency	Probability
F	F	F	200	0.2
F	F	T	50	0.05
F	T	F	50	0.05
F	T	T	50	0.05
T	F	F	50	0.05
T	F	T	150	0.15
T	T	F	50	0.05
T	T	T	400	0.4

$P(\text{passing the course}) = P(\text{Pass}=\text{T}) = 0.05 + 0.05 + 0.15 + 0.4 = 0.65$

$P(\text{attending lecture}) = P(\text{Lecture}=\text{T}) = 0.05 + 0.05 + 0.05 + 0.4 = 0.55$

What is the probability of a student passing who attends the work groups?

15

How do we combine two probabilities?

Conditional Probability

Conditional probability of an event A is the probability that the event will occur given the knowledge that event B will occur.

Assume $P(B) > 0$, then $P(A|B)$, the conditional probability of A given B is defined as follows:

$$P(A|B) = \frac{P(A, B)}{P(B)}$$

Alternatively...

$$P(A, B) = P(A|B) * P(B)$$

Also called Product Rule

16

The **conditional probability** of an event is the probability that it will occur *conditional* on another event having occurred. For example, what is the probability that a student passes *given* that the student attended lectures? We can express this as a formula from which we will later derive Bayes theorem. The notation $P(A|B)$ means the probability of A given that B . The $P(A, B)$ means the probability of A and B occurring (the intersection of a Venn diagram).

Conditional Probability *Example*

$$P(A|B) = \frac{P(A, B)}{P(B)}$$



Event A is rolling a 6: $A = \{6\}$, so $P(\{6\}) = \frac{1}{6}$

Event B is rolling an even number: $B = \{2, 4, 6\} = P(\text{even}) = \frac{1}{2}$

Probability of A given B is

$$P(\{6\} | \text{Even}) = P(\{6, \{2,4,6\}\}) / P(\{2,4,6\}) = \frac{1}{6} / \frac{1}{2} = \frac{1}{3}$$

Let's start with a dice example. The numerator of the equation becomes the probability of rolling a 6 *and* rolling an even number is simply $\frac{1}{6}$ because, out of all 6 options, these two conditions are only true at the same time if a 6 is rolled. In the denominator, the probability of rolling an even number is $\frac{1}{2}$. Then we calculate that $(\frac{1}{6}) / (\frac{1}{2})$ is $\frac{1}{3}$.

Inference by Enumeration

Inferencing is calculating some useful quantity from a probability distribution.
We can do this through inference enumeration.

e.g. $\mathbf{P}(\text{Toothache}, \text{Catch}, \text{Cavity})$

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

For any proposition ϕ , sum the atomic events where it is true

$$P(\phi) = \sum_{\omega: \omega \models \phi} P(\omega)$$

18

Above is a table showing a particular probability distribution $\mathbf{P}$ for the events toothache, catch, and cavity. This table allows us to easily inference over conditional probabilities by enumerating the probabilities. Let's try an example.

Inference by Enumeration *Example*

e.g. $P(\text{Toothache}, \text{Catch}, \text{Cavity})$

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	.108	.012	.072	.008
$\neg$ <i>cavity</i>	.016	.064	.144	.576

$$P(\text{Toothache}) = 0.108 + 0.012 + 0.016 + 0.064 = 0.2$$

The probability of having a toothache is the sum of all events in which Toothache = True, which is 0.2. Notice that finding the probability of Toothache = False can be done by simple calculating $1 - 0.2 = 0.8$.

Inference by Enumeration *Example*

e.g. $P(\text{Toothache}, \text{Catch}, \text{Cavity})$

	<i>toothache</i>		$\neg \text{toothache}$	
	<i>catch</i>	$\neg \text{catch}$	<i>catch</i>	$\neg \text{catch}$
<i>cavity</i>	.108	.012	.072	.008
$\neg \text{cavity}$	.016	.064	.144	.576

$$P(\neg \text{Cavity} \mid \text{Toothache}) = \frac{P(\neg \text{Cavity}, \text{Toothache})}{P(\text{Toothache})} = \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = 0.4$$

0.108+0.012+0.016+0.064

Now we calculate a conditional probability problem which asks, what is the probability of *not* having a cavity given that toothache is true? We apply the definition we have by summing the probabilities for which cavity = false and toothache = true in the numerator and dividing by the probability of toothache = true.

Probability Distribution *Example Revisited*

WorkGroup	Lecture	Pass	Frequency	Probability
F	F	F	200	0.2
F	F	T	50	0.05
F	T	F	50	0.05
F	T	T	50	0.05
T	F	F	50	0.05
T	F	T	150	0.15
T	T	F	50	0.05
T	T	T	400	0.4

Exercise: What is the probability of a student passing who attends the work groups?

$$P(\text{Pass}|\text{WG}) = \frac{P(\text{Pass}, \text{WG})}{P(\text{WG})} = \frac{0.15 + 0.4}{0.05 + 0.15 + 0.05 + 0.4} = 0.85$$

21

Applied to the Intelligent Systems probability distribution, the probability of passing given that you attend the workgroups is 85%.

Part 2: Applying Probability

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How can we use conditional probability and Bayes theorem?

Inversion Problem

The **inversion problem** states that it's easy to express the probability of the consequence given the cause, but we're usually interested in the reverse.

e.g. we are interested in $P(\text{disease} | \text{symptom})$, but typically $P(\text{symptom} | \text{disease})$ is better known.

e.g. reliability of a test is expressed as $P(\text{testResult} | \text{disease})$.

e.g. $P(\text{spam} | \text{wordUse})$ is better known than $P(\text{wordUse} | \text{spam})$.

This occurs because, e.g. there are fewer diseases than symptoms, and only two classes of spam or $\neg$ spam.



Thomas Bayes

23

The **inversion problem** says that we usually have the reverse information of what we're interested in; we want to know the cause given the effect, not the other way around. Doctors that observe certain symptoms in a patient are interested in the probability of the patient having a certain disease, but it is better known which symptoms are likely given the knowledge of a specific disease. When it comes to spam filtering, it is much more useful to know the probability of an email being spam given the use of words, but we only have data on the opposite.

Bayes' Theorem

The numerator in the rule of conditional probability is commutative...

$$P(A|B) = \frac{P(A,B)}{P(B)} \longrightarrow \begin{aligned} &= P(B,A) \\ &= P(B|A) * P(A) \end{aligned}$$

Bayes Theorem says if we assume $P(A) > 0$, and $P(B) > 0$, then

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

24

The rule of conditional probability is sometimes inadequate because the information that we have does not fit into the formula (inversion problem). Mathematician and reverend Thomas Bayes realised that the numerator in the rule is commutative, meaning the probability of A and B is the same as the probability of B and A. $P(B,A)$ can then be rewritten as a conditional probability and replace the old numerator, giving us Bayes Theorem.

Bayes Theorem *Example* (with fictional numbers)

Suppose 1% of the population has a condition called C.

The reliability of the test to determine C is as follows:

- the probability that someone with C gets a **positive** result is 0.792
- the probability of a **negative** result for someone who doesn't have C is 0.904

Exercise: What is the probability of you having C given a positive test result?

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

25

Let's try an example with Bayes Theorem using C (made up numbers). The probability that someone with the virus gets a positive test result is 0.792, and the probability of a negative test result for someone who doesn't have C is 0.904. These numbers look pretty promising, right? So what is the probability of having C given a positive test result?

Bayes Theorem *Example* (with fictional numbers)

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

$$\begin{aligned} P(C|+) &= \frac{P(+|C) * P(C)}{P(+)} = \frac{0.792 * 0.01}{P(+|C) * P(C) + P(+|-C)P(-C)} \\ &= \frac{0.792 * 0.01}{0.792 * 0.01 + 0.096 * 0.99} \approx 0.077 \end{aligned}$$

26

Let's plug in the numbers. *C* means the person has Covid, *+* means the person received a positive test result. There is only a 7% probability that you really have Covid given that you received a positive result. For this specific example, the result is very shocking because the **prior probability** of someone in the population having Covid is only 1%.

This concept of prior probability can be really useful in your personal life as well. For example, imagine that an expecting mother is told that her genetic makeup means that her child is 3 times as likely to have a birth defect than the general population. Although this sounds terrifying, if the *prior probability* of this birth defect occurring is only 0.001, then the child has a 0.003 probability of having it.

Part 3: Bayesian Networks (PGMs)

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27

In this next section we will look at Bayesian networks, also called probabilistic graphical models.

The Problem with Probability

Probability requires a lot of data.

The # of combinations of pieces of evidence grows exponentially in the #of pieces of evidence, *e.g.* 20 symptoms have > 1 million combination

Possible trick:

- Assume conditional independence of all the pieces of evidence given each hypothesis
- PRO: significantly reduces number of probabilities
- CON: assumption typically wrong and dangerously close to inconsistency

28

Frequentist probability requires a lot of data, a lot of “experiments”, in order to generalise to a population. For example, 20 different symptoms have over a million possible combinations. One trick for reducing the amount of data required is to assume **conditional independence**. We will formally define this in a few slides.

Joint Probability *Example Revisited*

WG	L	Pass	Understand	Bored	Frequency	Probability
F	F	F	F	F		
F	F	F	F	T		
F	F	F	T	F		
F	F	F	T	T		
F	F	T	F	F		
F	F	T	F	T		
F	F	T	T	F		
...						

What to do if we want to calculate $P(w, l, p, u, b)$?

29

In this table we introduce “Understand”, which is T when a student understands the material, and “Bored”. Now the problem is that 1000 students will not provide enough data for the number of different conditional probabilities we’re interested in, because some rows may have a very low frequency.

How to combine 3+ probabilities

e.g. $P(A,B,C) = ?$

Recall the product rule $P(A,B) = P(A|B) * P(B)$

Let $B,C = Y$ such that $P(A,Y) = P(A|Y) * P(Y)$

$$= P(A|B,C) * P(B,C)$$

$$= P(A|B,C) * P(B|C) * P(C)$$

This is called the **chain rule**, $P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | x_1, \dots, x_{i-1})$

This is still complex:

$$P(W,L,P,U,B) = P(W|L,P,U,B) * P(L|P,U,B) * P(P|U,B) * P(U|B) * P(B)$$

30

The product rule says that $P(A,B) = P(A|B) * P(B)$. The problem with calculating more than two probabilities is that there are not enough variables. We can solve this problem by assigning a variable Y to be the joint probability of two events, in this case B,C occurring. Now we are left with $P(A,Y) = P(A|Y) * P(Y)$. We can then re-apply the product rule so that we only end up with the right hand side of the formula and end up with $P(A|B,C) * P(B|C) * P(C)$. This is useful because neither B nor C depend on A anymore.

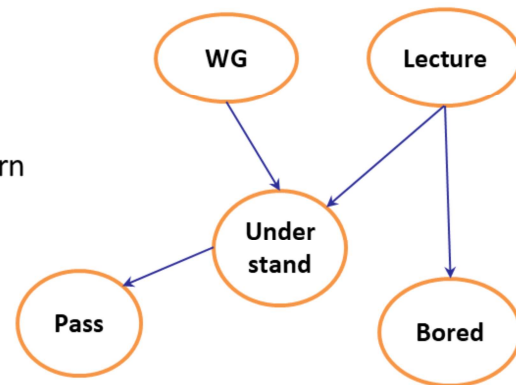
We can do these calculations for an n amount of random variables through the **chain rule**.

Bayesian Networks or probabilistic graphical models (PGMs)

Topology of network *encodes conditional independence*:

Understand and *Bored* are conditionally independent given *Lecture*.

Pass only depends on *Understand*, which in turn only depends on *WG* and *Lecture*



31

We can demonstrate these (in)dependencies by making random variables nodes and representing dependence with lines. In this graph, the weather is completely independent of each of the other variables. Toothache and catch are only independent given cavity.

Independence

Two events are **independent** when the occurrence of one does not affect the probability of the other.

If $P(A|B) = P(A)$, A is independent of B

Conditional independence says that, given knowledge of a third event C, knowing the occurrence of A doesn't affect the probability of B, and vice versa.

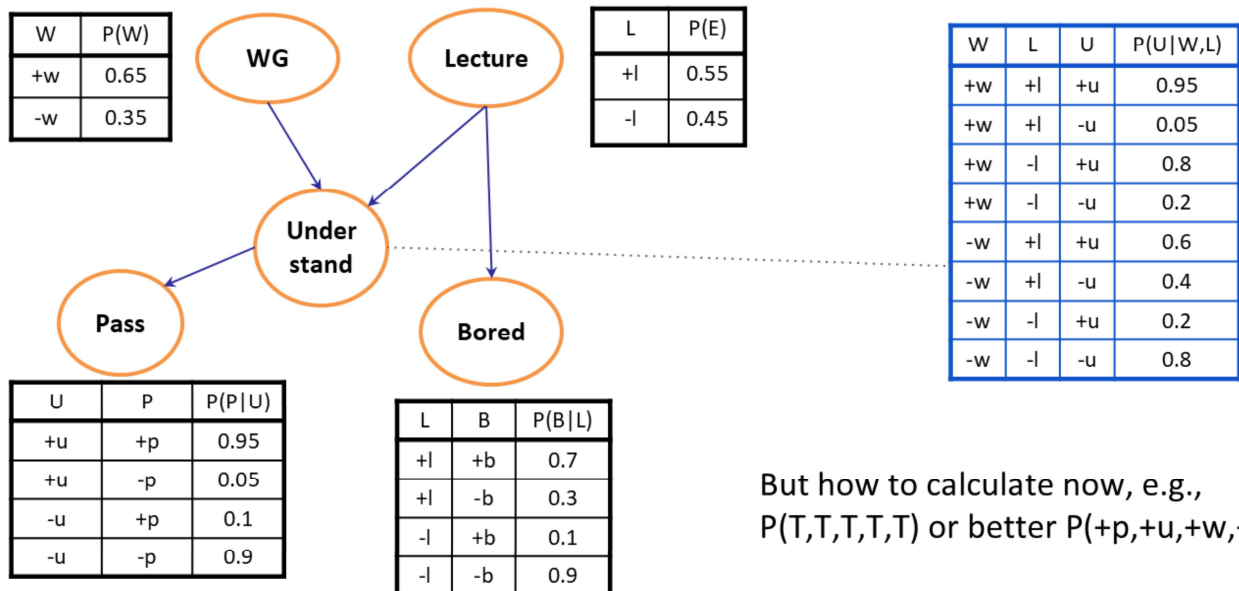
A is conditionally independent of B given C *iff*
 $A, B, C : P(A, B | C) = P(A | C) * P(B | C)$ or $A, B, C : P(A | C, B) = P(A | C)$

Exercise: Are events {1} and odd independent? Are odd and {1,2} independent?

32

Two variables are **independent** when the occurrence of one doesn't affect the other. We can formalise this by saying that if $P(A|B) = P(A)$, then A is independent of B. **Conditional independence** is useful when we're talking about independence amongst more than two random variables. The probability of A and B given C equals the probability of A given C times the probability of B given C. Otherwise said, the probability A given C and B is equal to the probability of A given C.

Bayesian Network Example



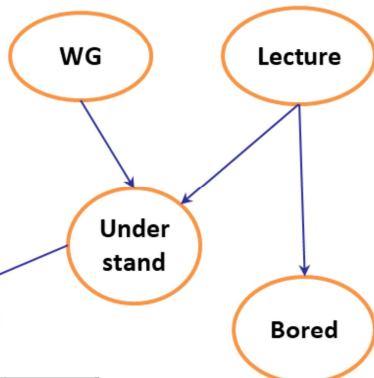
33

In this graph of the Intelligent Systems probability distribution, passing does not depend directly on attending the workgroups or lectures, but depends on whether you understand the material. We can make probability distributions for each random variable. For lecture and workgroup these tables are simple because they are independent of the other variables. However, understanding the material requires a table with every possible combination for work groups, lecture, and understanding. Note that a variable only depends on its direct parent node.

Instead of estimating the full 32 probabilities, we only have to do 20 now. The question remains on how to calculate the other values now.

Bayesian Network Example

W	P(W)
+w	0.65
-w	0.35



L	P(E)
+l	0.55
-l	0.45

Remember: $P(P,U,W,B,L) = P(P|U,W,B,L) * P(W|B,L) * P(B|L) * P(L)$
 Now we can remove non-dependent variables:

W	L	U	P(U W,L)
+w	+l	+u	0.95
+w	+l	-u	0.05
+w	-l	+u	0.8
+w	-l	-u	0.2
-w	+l	+u	0.6
-w	+l	-u	0.4
-w	-l	+u	0.2
-w	-l	-u	0.8

$$P(P,U,W,B,L) = P(P|U) * P(W) * P(B|L) * P(L)$$

And now we can calculate all the entries in our table.

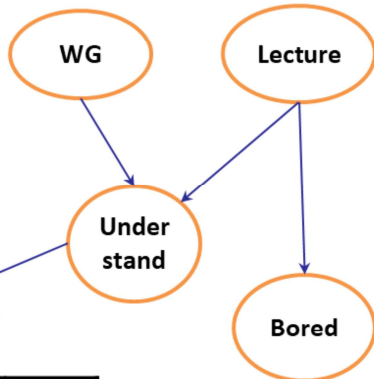
U	P	P(P U)
+u	+p	0.95
+u	-p	0.05
-u	+p	0.1
-u	-p	0.9

L	B	P(B L)
+l	+b	0.7
+l	-b	0.3
-l	+b	0.1
-l	-b	0.9

Recall that, with the chaining rule, the joint probability of all the variables relies on the dependence of all the variables on each other. By assuming conditional independence we can significantly shorten this formula by only writing down the dependence of the direct parent node of the variable.

Bayesian Network Example

W	P(W)
+w	0.65
-w	0.35



L	P(E)
+l	0.55
-l	0.45

What is the likelihood of students attending the work groups?

$$P(+w) = 0.65$$

U	P	P(P U)
+u	+p	0.95
+u	-p	0.05
-u	+p	0.1
-u	-p	0.9

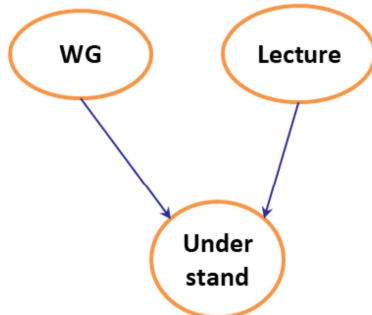
L	B	P(B L)
+l	+b	0.7
+l	-b	0.3
-l	+b	0.1
-l	-b	0.9

W	L	U	P(U W,L)
+w	+l	+u	0.95
+w	+l	-u	0.05
+w	-l	+u	0.8
+w	-l	-u	0.2
-w	+l	+u	0.6
-w	+l	-u	0.4
-w	-l	+u	0.2
-w	-l	-u	0.8

What is the likelihood of students attending the work groups?

Bayesian Network Example

W	P(W)
+w	0.65
-w	0.35



L	P(L)
+l	0.55
-l	0.45

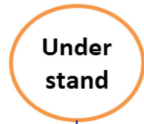
What is the likelihood of a student understanding something?
Depends on probability of WG and L.

W	L	U	P(U W,L)
+w	+l	+u	0.95
+w	+l	-u	0.05
+w	-l	+u	0.8
+w	-l	-u	0.2
-w	+l	+u	0.6
-w	+l	-u	0.4
-w	-l	+u	0.2
-w	-l	-u	0.8

$$P(+u) = P(+u, +w, +l) + P(+u, -w, +l) + P(+u, +w, -l) + P(+u, -w, -l)$$

$$P(+u, +w, +l) = P(+u | +w, +l) * P(+w) * P(+l) = 0.95 * 0.65 * 0.55 = 0.339$$

Bayesian Network Example



What is the likelihood of a student passing the course?
 Depends on Understanding, specifically probability of passing while understanding *plus* probability of passing while not understanding.

$$P(+p) = P(+p, +u) + P(+p, -u)$$

$$P(+p, +u) = P(+p | +u) * p(+u) = 0.95 * P(u+)$$
 RESULT FROM PREVIOUS

$$P(+p, -u) = P(+p | -u) * p(-u) = 0.1 * \text{RESULT FROM PREVIOUS}$$

U	P	P(P U)
+u	+p	0.95
+u	-p	0.05
-u	+p	0.1
-u	-p	0.9

W	L	U	P(U W,L)
+w	+l	+u	0.95
+w	+l	-u	0.05
+w	-l	+u	0.8
+w	-l	-u	0.2
-w	+l	+u	0.6
-w	+l	-u	0.4
-w	-l	+u	0.2
-w	-l	-u	0.8

Finally, calculating a variable that is dependent on one other variable can also be achieved through chaining and previously calculated totals.

Intelligent Systems Framework

Efficient, simple, correct, well- understood, generic	Find best decision based on experience.	Weeks 6 & 7
	Find best decision with knowledge & reason.	Weeks 3, 4, 5
	Find best decision in multiple agent setting.	Weeks 1 & 2
	Find best decision with heuristics.	
	Systematically find all possible solutions.	

Progress in the intelligent systems framework.

Summary

Take Home Message

We need the notion of **probability** to deal with **uncertainty**.

We can describe probability of **random variables** through sample and event spaces, and organise these probabilities into probability **distributions**.

We can view probability objectively or subjectively. For example, Bayes theorem which works with **conditional independence**.

Bayesian networks organise independences between events into space and are useful for calculating complex probabilities.

Thank you for your attention and have a great day.